



WEST BENGAL STATE UNIVERSITY
B.Sc. Minor/Cor 2nd Semester Examination, 2025

MTMMIN202T/MTMCOR202T-MATHEMATICS

Time Allotted: 2 Hours

Full Marks: 50

*The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.*

Answer Question No. 1 and any five from the rest

1. Answer any *five* questions from the following:

2×5 = 10

(a) Determine $f'(0)$, if it exists, for the function

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{for } x \neq 0 \\ 0, & \text{for } x = 0 \end{cases}$$

(b) Verify whether Cauchy's mean value theorem is applicable to the functions $f(x) = x^2$ and $g(x) = x^3$ on the interval $[1, 2]$.

(c) Evaluate: $\lim_{x \rightarrow \infty} \{x - \ln(x^2 + 1)\}$

(d) Evaluate: $\int x^3 e^x dx$

(e) Find length of the curve $x = \frac{1}{8}y^4 + \frac{1}{4}y^{-2}$ from $y=1$ to $y=4$.

(f) Find the radius of curvature of the curve $y = \log x$ at the point where $x=1$.

(g) Evaluate: $\int_0^{\pi/2} \frac{\sin^3 x}{\cos x} dx$

(h) Find $\frac{\partial f}{\partial y}$ and $\left(\frac{\partial f}{\partial y}\right)_{(1,0)}$ where $f(x, y) = x^3 y^2 + e^{xy^2}$.

2. (a) Does the limit $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\tan x} - 1}{e^{\tan x} + 1}$ exist?

3

(b) Find $\frac{dy}{dx}$ when $x^{\sin y} + y^{\cos x} = 1$.

5

3. (a) A function defined in the following way:

$$f(x) = \begin{cases} x, & \text{for } 0 < x < 1 \\ 2-x, & \text{for } 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2, & \text{for } x > 2 \end{cases}$$

Is $f(x)$ continuous at $x=1$? Is it differentiable at $x=2$? Show that $f'(1)$ does not exist.

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(b) If $y = e^{m \sin^{-1} x}$ show that

(i) $(1-x^2)y_2 - xy_1 - m^2y = 0$

(ii) $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (m^2 + n^2)y_n = 0.$

4. (a) If $V = z \tan^{-1} \frac{y}{x}$ prove that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0.$

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(b) Show that $\frac{x}{1+x} < \log(1+x) < x$, if $x > 0.$

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5. (a) Find the n -th derivative of $y = e^{2x} \cdot \sin 3x.$

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(b) If $u = \frac{x^4 + y^4}{x^2 + y^2}$, use Euler's theorem to find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}.$ ✓

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6. (a) Derive the reduction formula for $\int \sin^n x dx.$ ✓

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(b) Prove that: $\int_0^\infty e^{-x^2} x^7 dx = \frac{15}{8} \sqrt{\pi}$

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7. (a) Evaluate: $\int_0^1 \frac{x^2}{\sqrt{1-x^4}} dx$

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(b) Show that $\Gamma(n+1) = n\Gamma(n)$ and hence find $\Gamma(5/2).$ ✓

8. (a) Find the envelopes of the family of ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where the parameters are connected by $a+b=c$, c being constant.

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(b) Prove that the angle of intersection of the curves $x^2 - y^2 = p^2$ and $x^2 + y^2 = p^2 \sqrt{2}$ is independent of $p.$

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9. (a) Find the circle of curvature at t -point of the curve $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$, where t is a parameter.

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(b) Find the asymptotes of the curve:

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$$x^3 - 4xy^2 - 3x^2 + 12yx - 12y^2 + 8x + 2y + 4 = 0$$

—x—